

## Deployment scenario: real-time emotion monitoring of contact-center conversations

### Conversation stream

Customer: "I waited forty minutes."

Agent: "I apologize for the delay."

Customer: "Yeah, great service."

### Primary LLM (LLaMA2-7B)

per-utterance label + confidence

anger  $p = 0.91$   
sadness  $p = 0.86$   
joy  $p = 0.52$

**confidence gate**  
 $p \geq \tau$  ?

yes

73% of utterances: answer directly  
cost = 1× forward pass

no

27% of utterances: heterogeneous review  
second LLM (Qwen2.5-7B) re-classifies  
flip only if  $p_b > p_a + m$

escalated case: "Yeah, great service." (sarcasm)  
primary: joy (0.52) → critic: anger (0.78) → verdict: anger (error corrected)

expected cost = 1.27× forward passes

weighted-F1 68.6 → 69.6 (+1.0, seed 42; 8-seed mean +0.49,  $p=0.019$ )

zero-touch on 73% of traffic